

Light Chain Deposition Disease

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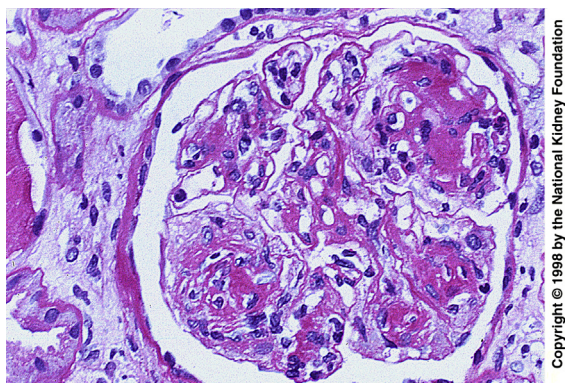


Fig 1. The characteristic appearance of light chain deposition disease by light microscopy is that of a nodular glomerulosclerosis, which strongly resembles diabetic nephropathy, as in this case (periodic acid-Schiff stain; original magnification $\times 400$).

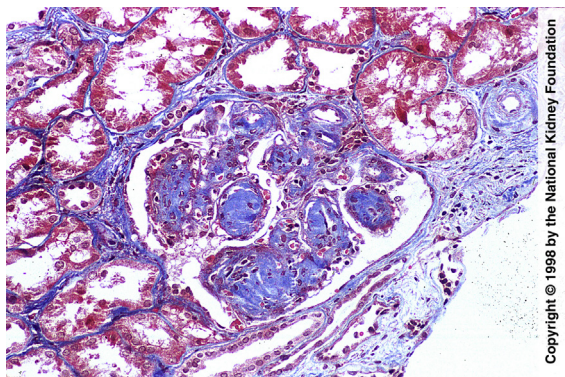


Fig 2. The typical nodular glomerulosclerosis appearance, which may be seen in light chain deposition disease, bears a striking resemblance to diabetic nephropathy, as in this case. Additional studies are necessary to differentiate between the two, although the minimal arteriolar hyalinization in this case in the presence of advanced nodular sclerosis suggests the possibility of a disease other than diabetes (Masson trichrome stain; original magnification $\times 200$).

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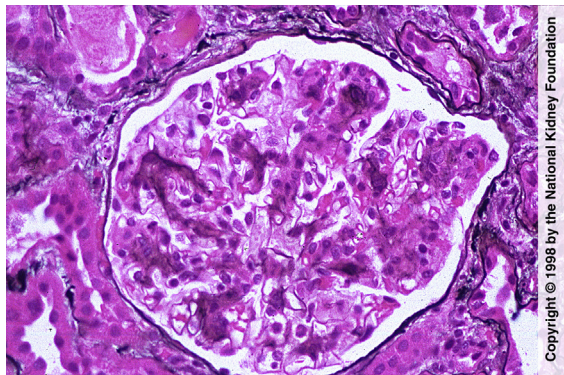


Fig 3. Although nodular glomerulosclerosis is the typical appearance of light chain deposition, minor degrees of mesangial expansion may be the only manifestation as in this case. There is mesangial matrix and cellularity increase without distinct nodule formation (periodic acid-Schiff stain; original magnification $\times 400$).

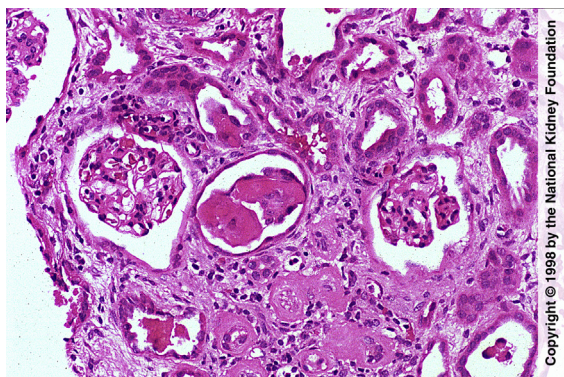


Fig 4. Only minor degrees of mesangial increase are present in this case, although additional studies by immunofluorescence and electron microscopy confirm that the patient had light chain deposition disease involving tubules and glomeruli, in addition to the characteristic myeloma cast nephropathy (illustrated in the center of the field). The characteristic appearance of myeloma cast nephropathy is illustrated by the presence of fractured, hard, refractile casts with surrounding syncytial cell reaction. Coexistence of two manifestations of paraprotein-related disease occurs, with a combination of myeloma and light chain deposition disease perhaps being the most common (hematoxylin & eosin stain; original magnification $\times 100$).

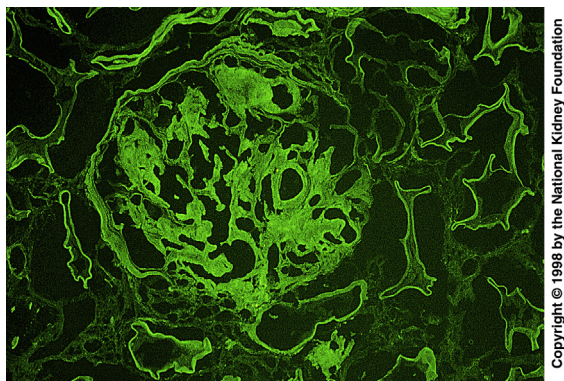


Fig 5. Glomerular capillary loop, mesangial staining, and linear tubular staining are characteristic of light chain deposition disease. Either kappa or lambda light chain paraprotein may cause light chain deposition disease, although kappa more commonly is the culprit (antibody to kappa light chain, immunofluorescence; original magnification $\times 200$).

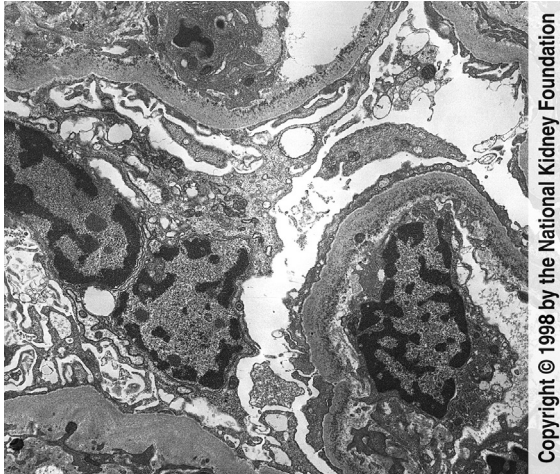


Fig 6. The granular, amorphous deposits typical of light chain deposition disease are seen as silt-like material on the endothelial aspects of the glomerular basement membrane in this case of light chain deposition disease (transmission electron microscopy; original magnification $\times 11,250$).

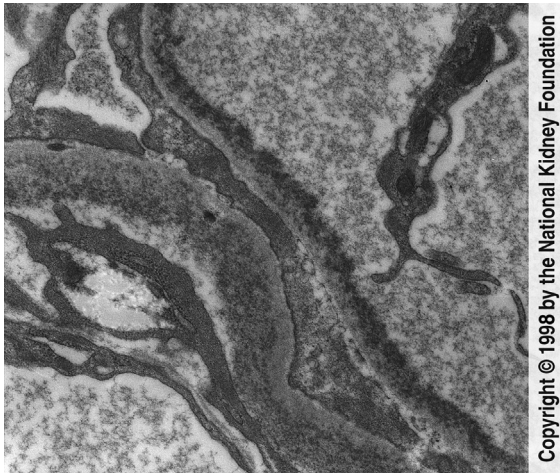


Fig 7. High-power view of the granular, amorphous deposits typical of light chain deposition disease, on the endothelial aspects of the glomerular basement membrane. These deposits have indistinct borders, are mottled, and occur both in the glomerular basement membranes and the mesangium and tubular basement membranes (transmission electron microscopy; original magnification $\times 40,000$). See other figures.

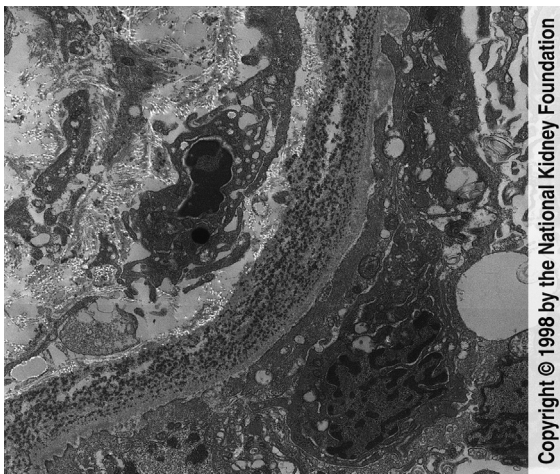


Fig 8. Tubular basement membrane with granular, amorphous deposits typical of light chain deposition disease (transmission electron microscopy; original magnification $\times 14,000$).